GCSE

Rewarding Learning

Chief Examiner's and Principal Moderator's Report

Digital Technology



Summer Series 2022

Foreword

This booklet outlines the performance of candidates in all aspects of this specification for the Summer 2022 series.

CCEA hopes that the Chief Examiner's and/or Principal Moderator's report(s) will be viewed as a helpful and constructive medium to further support teachers and the learning process.

This booklet forms part of the suite of support materials for the specification. Further materials are available from the specification's microsite on our website at www.ccea.org.uk.

Contents

Assessment Unit 1	Digital Technology Compulsory Core	3
Assessment Unit 2	Digital Authoring Concepts	5
Assessment Unit 3	Digital Authoring Practice	9
Assessment Unit 4	Digital Development Concepts	12
Assessment Unit 5	Digital Development Practice	16
Contact details		18

GCSE Digital Tehnology

Chief Examiner's Report

Assessment Unit 1 Digital Technology Compulsory Core

Unit Overview

Overall, the paper was well designed and permitted candidates to achieve good outcomes, it also catered for all levels of ability. Some outcomes were perhaps lower than expected and this may be due to some candidates taking the paper in their fourth year at Secondary School and their knowledge maybe somewhat limited and not used to external exams. The language within the paper was suitable for GCSE and the mark scheme provided enabled the markers to positively award candidates.

Q1 **MULTIPLE CHOICE**

This question was made up of parts a to h, each part contained four short statements and the candidate was asked to choose the correct response.

(a) - Well answered.
(b) - Well answered but some candidates wrote the number and not the letter.
(c) - Well answered but C was a common wrong answer.
(d) - Well answered.
(e) - Well answered.
(f) - Well answered.
(g) - Well answered.
(h) - Well answered.

Q2 This question was generally very well answered by the candidates, most were able to get two out of the three marks.

Q3 Question 3 was based on spreadsheets and was in the same format that the spreadsheet question was in the old GCSE ICT paper. While the easier parts of the question were well attempted it was somewhat disappointing that many candidates did not have the ability to understand the graph. This question was designed to determine that candidates can look at a graph which spreadsheets are generally designed to produce and to then be able to give the cell references used to create the graph. Many candidates could not answer the VLOOKUP aspect of the question. Understanding why a cell might contain an error message #N/A was not well answered.

Q4 Question 4 was well structured, and the majority of candidates scored well on this question, however many candidates do need to learn to write fuller answers.

Q5 This question was generally well answered but some candidates were unable to describe one advantages and one disadvantages of a 3D printer. This may indicate that the more difficult aspect of theory around system software are challenging to many candidates. Part 5(c) was not very well answered and it appeared candidates were unaware of the basic concept of a graphics digitiser.

Q6 Question 6 was the quality of written communication question. This question is designed to permit the examiner to assess the ability of the candidate to write coherently and to be able to discuss such topics. This question required the candidate to describe the disadvantages for customers shopping online. This question was in the main well attempted and many candidates were able clearly to describes sensible disadvantages. Many were able to write about half a page with some more able candidates writing three quarters of a page. Again, such a question can often be omitted by weaker candidates who do not like having to write detailed responses

Q7 Question 7 was well attempted by all candidates. Part 7(a) didn't seem to pose much difficulty to the majority of candidates. In 7(aii) candidates struggled somewhat in suggesting suitable disadvantages of a ring network and confused it with a star network. The rest of the question was very well answered and the majority of candidates got full arks on part 7(c).

Q8 Question 8(a) challenged many candidates when it came to explaining what is the Internet of Things? The rest of the question was well answered; however, part (d) was not as well answered as expected.

Q9 Question 9 was well attempted by all candidates, however some candidates found part (cii) where they were required in their own words to recall three principles of the data Protection Act difficult.

Q10 Question ten was the database question. The question was well structured and the candidates generally seemed to cope well with this question. This is possibly due to the coursework being based on databases.

Q11 This question was based on data portability and many candidates struggled to correctly state what is meant by Data Portability. Also, many struggled to expand the letters TIFF, which is a common file type. This was disappointing.

Q12 This question was partially answered well, candidates coped with expanding the letter WAN and explaining encryption. However, part (bii) and c were poorly answered. candidates did not clearly understand the impact the processor can have upon a computer's performance.

Assessment Unit 2 Digital Authoring Concepts

Unit Overview

In this unit, candidates develop their understanding of concepts associated with the development and use of digital multimedia systems. This constitutes 40% of the GCSE Digital Technology (Multimedia). The paper was worth 120 marks and the time allocated for completion was 90 minutes. Questions focused on the skills candidates developed across the unit and on a number of digital authoring concepts such as HTML and database and spreadsheet applications. In general, the paper provided facility for the majority of candidates to respond to all questions and discriminating questions were successful in providing differentiation. In evaluating candidate responses across the paper it was evident that a significant number of candidates are unable to provide appropriate definitions of key words or expand acronyms provided in the specification.

Q1 **(a)** A large proportion of candidates were unable to correctly identify what the letters https stood for.

(b) A significant number of candidates did not select the correct definition of the term feedback.

(c) A small proportion of candidates were able to correctly define the term composite key.

(d) This question was very well answered.

(e) The majority of candidates answered this question correctly.

Q2 **(a)** This question was well answered by the majority of candidates with a significant number of candidates getting full marks. A clear understanding of the key terms represented in this question was evident in the majority of candidate responses. Some candidates did on occasion confuse the terms Storyboard and Navigational Structure Diagram.

(b) Candidate responses to question parts (i) and (ii) indicated that there is still some confusion between the differences between various prototypes and their potential uses in the development of digital applications.

(c) Candidates were able to describe disadvantages of involving end users in prototype development with the majority citing issues around increased development time as being one of the major disadvantages. Students would benefit however from being reminded of the need to provide distinct examples in questions such as this one. A significant proportion of candidates provided two responses which were descriptions of the same issue and as a result, only one mark could be credited.

(d) This question, which focussed on the features of a good user requirements document was less well answered by candidates. A small number of candidates were awarded full marks for this question. Some candidates appeared to confuse target audience and user requirements and as a result focussed their answers around terms related to storyboarding and similar design tools focussing on developing a solution for a target audience as opposed to considering how best user requirements should be presented at the outset of a project.

(e) A large proportion of candidates were able to correctly identify 2-3 potential advantages of using HTML in website application development.

Q3 **(a)** In the main, responses to this question indicated candidates have a clear understanding of the key features of GUIs. A well answered question where the majority of candidates were awarded four or five out of five marks.

(b) Sensors and infra-red lights were common correct responses to this question. It was pleasing to see a large number of candidates responding well to this question with full marks being awarded in approximately 50% of cases. Common incorrect responses were cameras (without reference to infra-red).

(c) While the majority of candidates showed a clear understanding of the key features of touch-screen interfaces, marks were lost in some instances when consideration was not given to the context presented in the scenario. The scenario presented in this question related to a tourist information booth and as a result, responses which focussed on the portability feature associated with a touch screen interface would not be awarded marks.

(d) Most candidates were able to offer key disadvantages of voice data entry with some description of the disadvantages they had presented in their response. Candidates awarded higher marks for this QWC questions were those who were able to incorporate scenario related considerations into their responses.

Q4 Candidate responses to this question were very mixed. Some candidates referenced aspects of personal security in their response as opposed to considering the technical issues related to increased use of multimedia applications on mobile devices.

Q5 **(a)** The majority of candidates accessed marks in this question but a significant proportion of responses did show that some candidates did not have a clear understanding of the fact that each table in a relational database contains data about entities represented in the database.

(b) This question was not well answered by a significant number of candidates. Despite candidate experience in the development and use of relational database through the completion of controlled assessment tasks candidates had difficulty correctly identifying the types of relationships needed to link the tables presented in this scenario related questions.

(c) This question was well answered by the majority of questions showing a clear understanding of data types and their application in database development.

(d) Some candidates focussed their responses to this question on the password reference presented in the questions stem as opposed to fiving consideration to the types of validation checks which could be applied to a password in the situation presented. Where candidates directed their focus correctly to the question content this question was well answered by the majority.

(e) A significant number of candidates were able to correctly identify the fields/ tables required for two or three of the fields presented in this question. Some candidates referenced the need to include the GameID field, as opposed to GameName field in the query being designed. In these instances it was clear the candidate had an understanding of the question but had misinterpreted the data requirements of the query presented.

(f) A closed prose question relating to some of the key features of database applications. This questions was very well answered in the main, with most candidates being awarded full marks.

(g) Candidate response to this question indicated candidates had a clear understanding of the key features of macros with a significant number of candidates being able to correctly identify both correct statements presented in the table.

(h) Candidate responses to this question show there is an increased understanding of SQL syntax, especially straightforward SQL statements such as SELECT FROM. The majority of candidates were able to correctly complete the SELECT and FROM portions presented in question Part (i) and Part (ii) however the WHERE statement presented in Part (ii) proved to be more difficult and only a small number of candidates were able to access the final three marks in this question part.

Q6 **(a)** This question asked candidates to identify advantages of converting files to PDF format before sharing online. Responses were varied with a significant number of candidates correctly relating their responses to the reduction in download/ upload time or the reduction in web space taken up by the file in PDF format. Only a small number of candidates were awarded full marks for their response to this question.

(b) Responses to Part (i) and Part (ii) indicated candidates have a clear understanding of the meaning of the term optimisation with the majority of candidates being able to correctly identify optimised file types by assessing file extensions.

(c) In the main candidates were able to correctly identify at least one feature of a plug-in.

(d) The majority of candidates were able to correctly include one correct point in their definition of the term cross-platform compatibility. Responses tended to be non-technical however, showing students were responding based on experience, rather than using learned definitions.

(e) **(i)** Candidates tended to provide a description of how the <alt> tag operated on a web page, rather than linking their response to accessibility as per the question stem, as a result, this question part was not well answered.

(ii) Some candidates were able to draw from their controlled assessment experience to correctly answer this question and obtain full marks. A significant number of candidate's responses related to operating system accessibility tools as opposed to website accessibility features and as a result, this question part was also not well answered in the main.

Q7 **(a)** A proportion of candidates were not able to correctly identify what the letters HTML stood for.

(b) A proportion of candidates were not able to correctly identify what the letters CSS stood for.

(c) Candidate responses to this question indicated that the majority of candidates have a clear understanding of the structure and function of key HTML tags. This question was well answered by the majority of candidates.

(d) Candidate responses to this question showed that only a small number of candidates were familiar with the steps associated with event programming.

(e) Candidate responses to this question tended to include repetition of the stem here and examples of interactivity rather than explaining what the term interactive meant. Only a small number of candidates were awarded full marks to this question.

(f) This question was not well answered by the majority of candidates. While there is an increased understanding of the role played by scripting in website development, candidates are still not able to effectively describe the various methods of incorporating script into a HTML document.

(g) Candidates in the main were able to correctly identify the various script examples presented as either sequence in part (i) of this question. Some candidates did however confuse the terms repetition and selection in their responses to Part (ii) and Part (iii).

(h) & (i) Candidate responses to both question parts indicates a significant increase in student understanding of HTML code and their ability to determine how some HTML tags can be used to present content when viewing a HTML document. These questions were well answered by a significant proportion of candidates.

(j) A large proportion of candidates were not able to correctly identify what the letters URL stood for in response to Part (i) of this question. Very few candidates were able to correctly analyse all three HTML tags to determine if their associated functions were accurate. This was a difficult question and only a small number of candidates were awarded full marks.

Q8 **(a)** The majority of candidates were able to access some of the marks available in this question which focussed on different approaches to testing digital applications. Only a small number of candidates were able to access all 6 marks however.

(b) Some students misinterpreted this question and responded by addressing the various elements of a multimedia application which should be tested. Where candidates correctly identified the need to reference types of test data needed in a test plan, the majority were able to access three or four of the four marks available in this question.

Q9 **(a)** Candidate responses to this question indicate there is a clear understanding amongst candidates around the process of evaluation, who is involved in software evaluation and the purpose of evaluation. A large number of candidates are not aware of the ongoing nature of evaluation.

(b) This was difficult question with only a small number of candidates being able to correctly identify the reason for each type of test in addition to being able to give an appropriate example from the information provided in the question stem. Where candidates accessed all six marks available in this question, they were able to use examples from the question stem to explain how evaluation could help ensure a complex, robust and efficient solution is developed.

Q10 Only a small number of candidates were able to correctly match the definitions provided to the correct key term from the question stem.

Principal Moderator's Report

Assessment Unit 3 Digital Authoring Practice

Controlled Assessment Tasks

All centres successfully submitted controlled assessment tasks electronically. Centres must ensure candidates have been provided with the correct scenario for Units 3. This information is clearly provided on the front cover of the controlled assessment units. Controlled assessment guidelines must be followed throughout tasks and 'modelling' using live data files is not authorised.

Administration

The following procedures must be adhered to:

- All candidates of the requested sample, including the top and complete bottom candidate of the sample must be included.
- Only essential files should be submitted for moderation. Original data files or duplicate word processing files that have been converted to Portable Document Format should not be included. A small number of centres did not meet this particular requirement.

Accuracy of marking

The accuracy of assessment judgements from centres was generally very high and demonstrated a sound understanding of the controlled assessment requirements. A small number of centres marked leniently and marking was adjusted. In some cases, marking was slightly harsh at the lower end of the mark range. Centres should use the 'best fit' mark band approach. The use of templates is prohibited. Candidates are encouraged to produce individual solutions. Assessment of testing and evaluative sections was an area of concern for some centres.

The majority of centres correctly applied the mark scheme and the banding approach to the assessment of candidate work. It is imperative that all centres complete internal standardisation effectively before submitting marks to CCEA. Some centres did not follow the mark scheme and/or adhere to internal standardisation procedures.

It was positive to note the range of techniques used by candidates in the completion of controlled assessment tasks. There is no requirement for candidates to produce screenshots or step by step reports for controlled assessment tasks. Thank you to all centres for their efforts this series with electronic controlled assessment submission.

Task 1 Designing the multimedia and database solution using appropriate tools

Many examples of good practice have been noted in the completion of Task 1 by candidates. The following points should be noted:

- The majority of storyboards were computer generated. If 'hand-drawn' storyboards are used, they must be scanned to a high resolution to assist the moderation process. Rather than the focus on the quantity of storyboarding, candidates should focus on the quality and detail of storyboards, for example, the design of the website could include the design of the masterpage. Database design may include a more text/ tabular form/query/report design, to reduce the duplication of work.

- One example of each query, form, report planning is a requirement. More detailed planning on a masterpage would reduce workload, yet still a produce a high quality multimedia solution.
- Storyboards must show the sources of information and file types to be used.
- The design of professional reports is to be commended, with full titles and the company logo applied appropriately.
- However, many candidates failed to plan complex queries (two or more criteria) or appropriate reports (grouping, sorting and calculations).
- Some candidates did not attempt prototyping and the use of feedback in refining the solution, this section was marked leniently. 'Marking up' of prototype amendments and feedback could be applied to storyboards in a different font colour.
- Some centres produced unnecessary detail in their design sections e.g. full prototype designs, followed by full storyboards.

Task 2 Building the multimedia and database solution

Similarly, many examples of good practice and unique approaches to the task have been noted in the completion of Task 2 by candidates. The following points should be noted:

- Graphics, animation and videos produced for the candidate websites were generally of a high standard.
- Overall the standard of web authoring was high. It is positive to note many candidates using masterpages and cascading style sheets.
- Scripted elements may include: form validation, slideshow, animation, login, CSS, colour (hex).
- Accessibility elements may include: browser settings, plugins (downloads), speech, magnification, Alt tags.
- The location of scripted and accessibility elements must be noted on the eCandidate record sheets.
- Candidates should be encouraged to name their website homepage as 'index' or 'home'.
- Database solutions incorporated appropriate relationships, user friendly and intuitive switchboards, in addition to appropriate validation and input masks.
- Database solutions were implemented from the design document, however the depth of complex queries and reports (grouping, sorting and calculations) were lacking, particularly grouping and calculations.

Task 3 Testing the database solution

- Testing must include a range of valid, invalid and extreme data.
- Some candidates designed excessive tests and included screenshots for all testing cases, this could be further refined.
- This section was marked leniently by some centres.
- No screenshots are required for testing; screenshots are only required to demonstrate corrective action.

Task 4 Evaluating the multimedia and database solution

- Some candidates described their solution, rather than develop evaluative language. A summary of strengths, weaknesses and improvements led to descriptive rather than evaluative comments.
- It is recommended that candidates structure their evaluation logically in terms of the extent to which the user requirements have been met: fully met, partially met or not met.
- Top candidates successfully addressed performance and robustness issues, with refinements/improvements clearly identified. Performance, robustness and refinements were not always given in depth consideration.
- This section was marked leniently by some centres.

Chief Examiner's Report

Assessment Unit 4 Digital Development Concepts

Unit Overview

In this unit, candidates in this unit, students analyse trends in software development and the concepts involved in designing and building digital systems using coded solutions. Candidates are encouraged to investigate programming structures and the associated concepts to enable the creation of algorithmic solutions as part of the design process when creating programmes.

The unit constitutes 40% of the GCSE Digital Technology (Programming). Candidates had 90 minutes to complete a paper with 120 marks allocated. The paper focused on the candidates holistic understanding of programming concepts and structures. For example, candidates were required to design algorithms and flow charts to represent solutions to problems posed. This meant candidates needed assigned understanding of sequence, selection, and iteration.

Overall, the paper successfully provided opportunities for the full range of candidates to demonstrate their knowledge and understanding in this area. The significant majority of candidates were able to answer simple questions about programme design and data types. Many candidates scored well in the section on number systems and on closed response questions. After considering candidate responses it was clear that a significant number of candidates were unable to describe fully the bubble sort and the insertion sort. In addition, questions involving data structures, such as Question 7, proved to be difficult for some.

Q1 **(a) & (b)** About truth tables. This was well answered by most candidates with many achieving two marks for their responses.

(c) This question tests the candidates' ability to predict the outcome from sections of algorithms given values. Many answered both parts correctly. The loop part of the question proved more challenging and candidates in the lower bands did not answer correctly. Many selected D which left out the zero from the initial multiplication of the loop counter.

(d) & (e) Related to computational thinking was well answered and candidates achieved the full marks in many cases.

(f) A question on search techniques was less well answered as some candidates were unable to distinguish between a linear and binary search.

(g) Acronym expansion – generally well-answered by all candidates across the range.

Q2 This is the first and most simple of the algorithmic solution questions. candidates are asked to complete a simple exercise using sequence and selection, loops are not required.

(a) Most candidates correctly identified an acceptable data type.

(b) This required candidates to complete the algorithm. Candidates were required to have formulated their conditions correctly. Conditions using nested if statement were also acceptable provided the correct output was achieved. A considerable number of candidates entered incorrect symbols for "less than" and "greater than".

(c) **(i)** This question was designed to assess the candidates understanding of test data. The majority of candidates were able to identify tests for valid, invalid and extreme data. Many were also able to identify the test for accurate output.

(ii) Again, this was well answered. A high proportion of candidates provided a correct value less than 30 or referred to the use of data of a different data type or a negative value. All of these were acceptable.

Q3 **(a)** This was a closed response question relating to the translation process. The access ability of this question was evidenced by the number of students achieving marks in the top band. Candidates' responses demonstrated that they understand the need for the translation of high-level programming code

(b) This question related to the features of a software development environment. It was pleasing to see that candidate responses showed detailed knowledge of the relative programming tools used across schools and colleges. They were able to identify the key features of software development environments. Again, many achieved full marks.

Q4 Question 4 required candidates to write and complete algorithmic solutions. This question was more difficult than question two as candidates were required to understand and predict the outcome from sequence selection and iteration.

(a) Candidates were presented with a table of values from which they were required to construct an algorithm to calculate the cost of cupcakes ordered. It should be noted that the question states that all if statements used should contain only one condition. Candidates were therefore required to construct nested if statements in order to solve the problem. Candidates were allocated marks for using the correct conditions and allocating the correct price based on the conditions. Several different solutions were acceptable provided the output achieved from the solution was correct. Candidates in the higher level of achievement were able to complete this task. A significant majority of these candidates achieved full marks. Some candidates were unable to construct the algorithm using only one condition in each of statements and whilst these candidates made a good attempt at assigning the correct price for the cupcakes, they lost marks due to the composition of the conditions. Candidates in both the middle and higher level and should be able to construct nested if-statements from a table such as the one presented in Part (a).

(b) **(i)** This part of the question required candidates to complete an algorithm which related to the validation of a number. Candidates performed well constructing the condition in the if statement however, the assignment of the Boolean value valid in the algorithm caught many candidates out. A sizeable number set valid equal to true before out putting the error message. Candidates should be able to write validation algorithms using Boolean variables to control the loop structure in which they sit.

(ii) The question on data types was very well answered by the majority of candidates, indicating a strong understanding of these concepts from within the specification.

(c) Related to the processing of strings. in the first part of this question a significant number of candidates indicated their understanding of string concatenation using the '+' symbol to add ErrorMessage to ErrorMessage2 to giving ErrorMessageAll. This was an acceptable answer. Candidates providing an acceptable and coherent explanation, of string concatenation, were also given credit. Responses indicated that candidates understand how to process strings through concatenation and the length function.

Q5 This question tested the quality of written communication. The question related to sorting and candidates were required to describe how each sort method works and to comment on the efficiency of it. The majority of candidates were able to describe aspects of the bubble sort for example, comparisons and passes. A lesser number

were able to describe, accurately, the features of the insertion sort. In addition, whilst candidates where able to refer to the efficiency of both, answers relating to the insertion sort where poorly articulated and in many cases incorrect. At this level, candidates should be able to provide a simple, but clear, explanation of how each of the sorts work and their relative efficiency.

Q6 This question related to number systems and the conversion of numbers from denary to binary and hexadecimal. All parts of the question where well attempted. a minority of candidates did not refer to the value provided in Part (b)(i) when giving the explanation of a nibble, therefore they could only access one of the marks. Candidates should be reminded to read the question carefully and where they are directed to use a value provided, they should do so.

In Part (c) of the question candidates demonstrated, very well, their understanding of overflow and adding two bytes together. This part of the question was well answered, and a significant number of candidates achieved full marks.

Part (d) of the question was designed to test candidates' knowledge and understanding of storage units. Candidates were presented with a file size and a hard disc size and asked to work out if the file would fit onto the hard disc. The mark scheme was expanded to accept both 1024 and 1000 as the multiplier. Many candidates achieved full marks. Some candidates lost a mark as they did not show all their working out, clearly. Candidates should be encouraged to read instructions carefully and follow the instructions as outlined on the examination paper.

Q7 This question related to the use of the list or array data structure. Again, the demand is increasing from Question 4. In this question candidates are required to display understanding of the mechanics of array processing.

(a) **(i)** Many candidates were able to explain that the third value in the array traffic was equal to 236. However, a minority of candidates expressed this as a statement e.g., traffic[2]. Candidates are expected to know that an individual element within an array or list structure is accessed using the array or list name followed by the position or index of the item in the array or list. A suitable example would gain a further mark. A significant number of candidates achieved only two marks in this part of the question as they did not make use of an example to illustrate their answer.

(ii) This part of the question provided some stretch and challenge for candidates. Candidates were required to construct a loop, process an array within the loop, maintain a running total and calculate several values based on the calculations performed inside the loop. Marks were awarded on a unitary basis for the inclusion of any or all of the aspects requested in the question. This question, set to differentiate, achieved its objective. There was a spread of marks allocated with many candidates scoring in the range 5 to 8. Candidate responses indicated a good knowledge of loops, assignment statements, an understanding of running totals and to a lesser extent how to process an array using the loop counter.

(b) Candidates were required to describe what a function/procedure is and in the second part to state 2 benefits of using user-defined functions in programme code. The first part of the question was well answered. Many candidates referred to a named section of code written to perform a specific task. Fewer candidates referred to the fact that a function/procedure can return a value or that it can have parameters. In the second part of the question, most candidates identified the advantage of re-usability along with reference to improved testing. Some candidates referred to helping programmers to understand the code, this

was not an acceptable answer however, improved code readability was accepted. Candidates should understand the difference between these two responses and why the former is not acceptable.

Q8 (a) Was designed to assess the candidates' knowledge and understanding of flow charts as an algorithmic solution. Part (a) was well attempted by the majority of candidates. Key areas where candidates went wrong include: the construction of the condition in the decision box and the positioning of the arrows to indicate the loop. it was clear, that candidates had a good understanding of the symbols related to flow charting but again the construction of conditions using inequalities and some logic proved challenging for some.

(b) Part (b) of the question related to a dry run. This part of the question was not well answered. Candidates were unable to accurately articulate the nature of a dry run and its function nor were they able to explain how a programmer would create a dry run. Being able to construct and execute a dry run will help students predict the outcome from code or algorithms and will develop their computational thinking skills. This activity is to be encouraged and could be utilised in the development of their code in the controlled assessment task.

Q9 (a) This question about file handling functions was well answered. Most candidates were able to name a file handling function and explain its purpose. A small number of candidates used read and write functions and therefore achieved no marks, since the question stem excluded these.

(b) Again, this part of the question, relating to the different types of errors, was very well answered, indicating that candidates are making effective use of practical programming and understand how these errors are caused. Although candidates scored well in this part of the question, in many cases examples were not well chosen. For example, candidates commonly wrote a statement of code with a spelling error or missing symbol but did not explain it. Marks were allocated in the majority of cases, but candidates are to be encouraged to explain how their chosen examples cause the error identified.

Q10 (a) A closed response question answered well by the majority of candidates.

(b) (i) & (ii) These questions were related two error trapping. a significant number of candidates confused this concept with validation. There is a distinct difference between validation and error trapping. Candidates at this level should be able to make the distinction and provide suitable simple examples of how their chosen coding language provides for error trapping. The mark scheme was expanded. Acceptable responses included Try Catch and exception handling. However simpler answers which referenced stopping crashing or mentioned runtime were also acceptable this year. Very few candidates referred to exception handling.

Q11 (a) A closed response question relating to evaluation which was answered well by most candidates.

(b) In this part of the question candidates were required to apply their knowledge of evaluation to a scenario. Candidates were provided with different forms of evaluation and were asked to explain their importance and to give an example from the information provided. In the main candidates were able to explain the importance of the different types of evaluation but few were able to provide suitable valid examples from the information provided. Candidates achieving full marks were able to provide solid examples from within the scenario.

Principal Moderator's Report

Assessment Unit 5 Digital Development Practice

Controlled Assessment Tasks

All centres successfully submitted controlled assessment tasks electronically. Centres **must** ensure candidates have been provided with the correct scenario for Unit 5. This information is clearly provided on the front cover of the controlled assessment units. Controlled assessment guidelines must be followed throughout tasks and 'modelling' using live data files is not authorised.

Administration

The following procedures must be adhered to:

- All candidates of the requested sample, including the top and complete bottom candidate of the sample must be included.
- Only **essential files** should be submitted for moderation. Original data files that have been converted to **Portable Document Format** should not be included. A small number of centres did not meet this particular requirement.

Accuracy of marking

The accuracy of assessment judgements from centres was generally very high and demonstrated a sound understanding of the controlled assessment requirements. A small number of centres marked leniently and marking was adjusted. The use of templates is prohibited. Candidates are encouraged to produce individual solutions. Assessment of testing and evaluative sections was an area of concern for some centres.

The majority of centres correctly applied the mark scheme and the banding approach to the assessment of candidate work. It is imperative that **all** centres complete internal standardisation effectively before submitting marks to CCEA. Some centres did not follow the mark scheme and/or adhere to internal standardisation procedures.

It was positive to note the range of techniques used by candidates in the completion of controlled assessment tasks. There is no requirement for candidates to produce screenshots or step by step reports for controlled assessment tasks. Thank you to all centres for their efforts this series with electronic controlled assessment submission.

Task 1 Design a solution using appropriate tools

Many examples of good practice have been noted in the completion of Task 1 by candidates. The following points should be noted:

- In some instances, user requirements were extracted from the task, without development or interpretation. Target audience needs must also be fully identified.
- The design component was particularly well implemented with high quality flowcharts and pseudo code included effectively.
- Validation and error trapping techniques were designed appropriately, however use of user feedback to refine the solution was omitted by a number of centres and marked leniently.

Task 2 Building a solution

Similarly, many examples of good practice and unique approaches to the task have been noted in the completion of Task 2 by candidates. Centres are to be highly commended upon their approach to this task. The following points should be noted:

- **Program usernames and passwords must be noted on the eCandidate record sheets. A number of centres are not adhering to this essential element of controlled assessment submission.**
- Marking was highly accurate across the sampled centres and the programming solutions were of a high standard.
- Candidates should be encouraged to develop their own program style and content.
- Solutions of the highest standard gave due consideration to system robustness.
- All solutions must be annotated.

Task 3 Testing the solution

- Generally, test plans were well structured, in tabular format (showing tests, expected outcome, actual outcome and fixes), in some cases test plans were cluttered and could be more professionally presented.
- It is very important that testing incorporates a range of tests (black and white box testing) and includes valid, invalid and extreme test data. Extreme testing of data was lacking in some centres.
- Some candidates designed excessive tests and included screenshots for all testing cases, this could be further refined.
- **No screenshots are required for testing; screenshots are only required to demonstrate corrective action.**

Task 4 Evaluating the solution

- Some candidates described their solution, rather than develop evaluative language. A summary of strengths, weaknesses and improvements led to descriptive rather than evaluative comments.
- It is recommended that candidates structure their evaluation logically in terms of the extent to which the user requirements have been met: fully met, partially met or not met.
- Top candidates successfully addressed performance and robustness issues, with refinements/improvements clearly identified. Performance, robustness and refinements were not always given in depth consideration.
- This section was marked accurately by most centres, however a small number of centres marked performance, robustness and refinements section leniently.

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